Traffic Simulator Statistics Summary

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Below are the statistics summaries for 3 different kinds of junctions:

1) Basic Junction:
    - Randomly picks 1st waiting vehicle from incoming road when intersection empty.
    - Once vehicle enters, 0.3-second delay applied and junction is considered occupied- restricts other vehicles from entering until active car successfully transfers to outgoing road, remaining trapped and blocking the entire junction if the destination lane is full.
2) Traffic Signal Junction:
    - Monitors incoming roads, allocates 5.0s green light to lanes with active queues.
    - Once vehicle enters from green-lit road into intersection, 0.3-second delay applied and junction is considered occupied- restricts other vehicles from entering until active car is transferred to outgoing road based on its destination.
3) Roundabout Junction:
    - Models circular roundabout with finite capacity. Junction manages vehicles already inside the circle and admits new ones.
    - Once a vehicle's timer reaches zero inside the circle, it is routed onto its designated exit road and removed from roundabout's active pool.
    - If roundabout is below maximum vehicle capacity, it randomly selects a waiting vehicle (FIFO for each incoming road) from all incoming roads to enter.
    - Upon entry, entry and exit angles based on physical geometry found to determine customized travel time proportional to this angular distance (full loop is 3.0s, with 0.5s minimum).

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SIMULATION STATISTIC SUMMARY

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Time Step (dt): 0.1 s
Total Steps: 500
Network Size: 7 Nodes, 9 Roads
[Traffic Generation Rates]
SRC1 : 2 vehicles/sec
SRC2 : 4 vehicles/sec
Total Time Simulated: 49.9 s

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For Basic Junction:

Total Cars Completed: 128
Exact Average Wait Time: 1.28 s per car
Network Average Throughput: 2.565 cars/sec
Throughput by Destination:
SNK1 : 49 cars (0.982 cars/sec)
SNK2 : 79 cars (1.583 cars/sec)

Queue Lengths per Road:
R1 : Avg 1.11 cars | Max 11 cars
R2 : Avg 0.29 cars | Max 3 cars

R3 : Avg 0.02 cars | Max 1 cars
R4 : Avg 0.79 cars | Max 5 cars
R5 : Avg 0.83 cars | Max 6 cars
R6 : Avg 0.78 cars | Max 6 cars
R7 : Avg 0.78 cars | Max 8 cars
R8 : Avg 0.00 cars | Max 0 cars
R9 : Avg 0.00 cars | Max 0 cars

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For Traffic Signal Junction:

Total Cars Completed: 127
Exact Average Wait Time: 1.01 s per car
Network Average Throughput: 2.545 cars/sec
Throughput by Destination:
SNK1 : 47 cars (0.942 cars/sec)
SNK2 : 80 cars (1.603 cars/sec)

Queue Lengths per Road:
R1 : Avg 0.18 cars | Max 3 cars
R2 : Avg 0.14 cars | Max 2 cars
R3 : Avg 0.01 cars | Max 1 cars
R4 : Avg 1.45 cars | Max 8 cars
R5 : Avg 0.12 cars | Max 2 cars
R6 : Avg 0.46 cars | Max 4 cars
R7 : Avg 0.64 cars | Max 6 cars
R8 : Avg 0.00 cars | Max 0 cars
R9 : Avg 0.00 cars | Max 0 cars

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For Roundabout Junction:

Total Cars Completed: 140
Exact Average Wait Time: 0.17 s per car
Network Average Throughput: 2.806 cars/sec
Throughput by Destination:
SNK1 : 60 cars (1.202 cars/sec)
SNK2 : 80 cars (1.603 cars/sec)

Queue Lengths per Road:
R1 : Avg 0.01 cars | Max 2 cars
R2 : Avg 0.00 cars | Max 1 cars
R3 : Avg 0.02 cars | Max 1 cars
R4 : Avg 0.00 cars | Max 0 cars
R5 : Avg 0.00 cars | Max 1 cars
R6 : Avg 0.08 cars | Max 2 cars
R7 : Avg 0.03 cars | Max 2 cars
R8 : Avg 0.00 cars | Max 0 cars
R9 : Avg 0.00 cars | Max 0 cars

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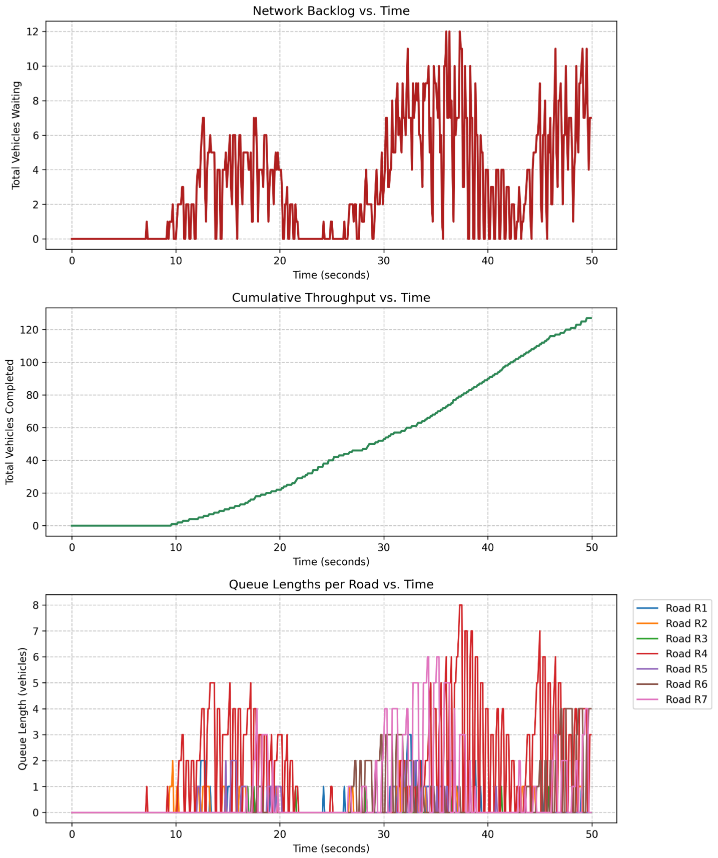
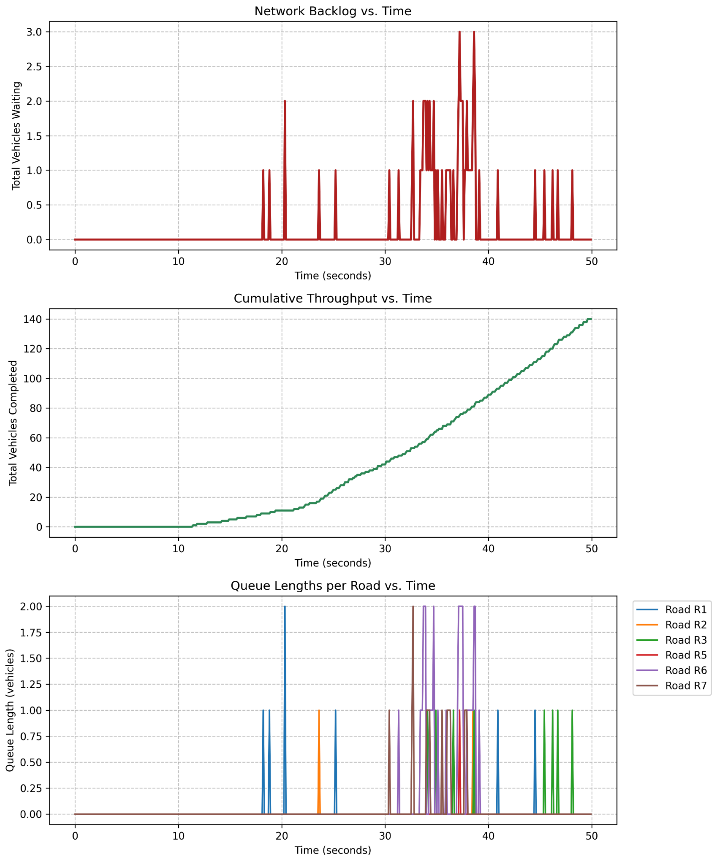
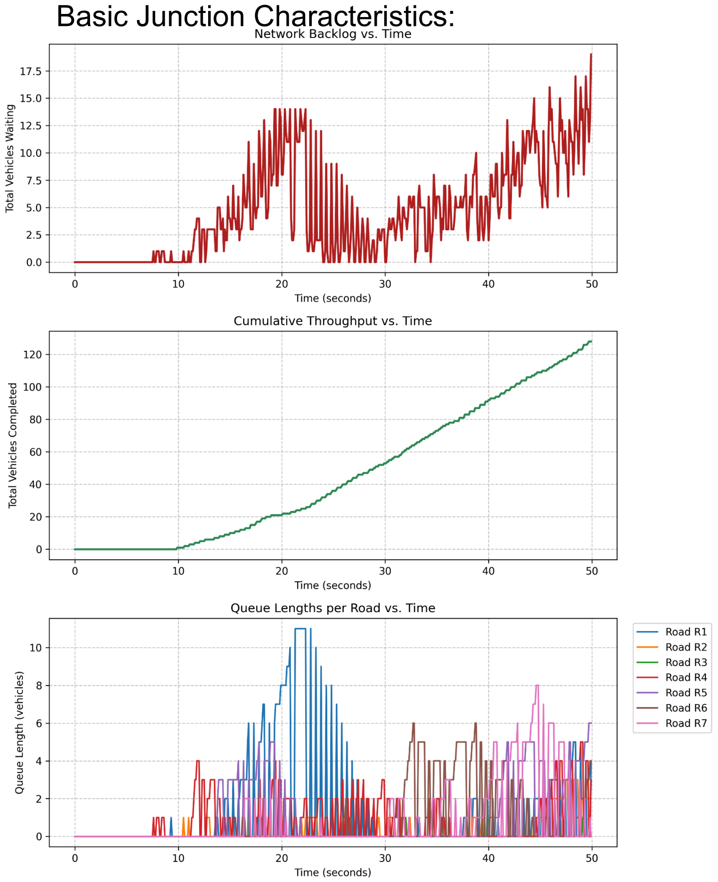
Traffic Signal Junction Statistics:



Roundabout Signal Junction Statistics:



Basic Junction Characteristics:



Roundabout Junction maximizes overall throughput (140 cars) with a wait time of just 0.17s/car, given its higher capacity. The Basic Junction and Traffic Signal Junction process a nearly identical volume of traffic (~127-128 cars), but the smart signals give an advantage in queue management.

While the uncoordinated Basic Junction struggles with localized bottlenecking- seen in the severe 11-car maximum queue on R1 and a higher wait time of 1.28 seconds, the Traffic Signal successfully mitigates this, reducing the R1 queue to just 3 cars, lowering the overall network wait time to 1.01s.

While signalling more evenly distributes waiting vehicles compared to the basic approach, the higher-capacity geometry of the roundabout is more efficient at handling this network.